

Overview of LLaMA language model

Subject: LLaMA language model

1.

Llama 1 models are only available as foundational models with self-supervised learning and without fine-tuning. Llama 2 – Chat models were derived from foundational Llama 2 models. Unlike GPT-4 which increased context length during fine-tuning, Llama 2 and Code Llama - Chat have the same context length of 4K tokens. Supervised fine-tuning used an autoregressive loss function with token loss on user prompts zeroed out. The batch size was 64. For AI alignment, human annotators wrote prompts and then compared two model outputs (a binary protocol), giving confidence levels and separate safety labels with veto power. Two separate reward models were trained from these preferences for safety and helpfulness using reinforcement learning from human feedback (RLHF). A major technical contribution is the departure from the exclusive use of proximal policy optimization (PPO) for RLHF – a new technique based on rejection sampling was used, followed by PPO. Multi-turn consistency in dialogs was targeted for improvement, to make sure that "system messages" (initial instructions, such as "speak in French" and "act like Napoleon") are respected during the dialog. This was accomplished using the new "Ghost attention" technique during training, which concatenates relevant instructions to each new user message but zeros out the loss function for tokens in the prompt (earlier parts of the dialog).

2.

Software developer Georgi Gerganov released llama.cpp as open-source on March 10, 2023. It's a re-implementation of Llama in C++, allowing systems without a powerful GPU to run the model locally. The llama.cpp project introduced the GGUF file format, a binary format that stores both tensors and metadata. The format focuses on supporting different quantization types, which can reduce memory usage, and increase speed at the expense of lower model precision. llamafile created by Justine Tunney is an open-source tool that bundles llama.cpp with the model into a single executable file. Tunney et al. introduced new optimized matrix multiplication kernels for x86 and ARM CPUs, improving prompt evaluation performance for FP16 and 8-bit quantized data types.

3.

Training datasets Llama's developers focused their effort on scaling the model's performance by increasing the volume of training data, rather than the number of parameters, reasoning that the dominating cost for LLMs is from doing inference on the trained model rather than the computational cost of the training process. Llama 1 foundational models were trained on a data set with 1.4 trillion tokens, drawn from publicly available data sources, including:

4.

Initial release The first version of Llama (stylized as LLaMA and sometimes referred to as Llama 1) was announced on February 24, 2023, via a blog post and a paper describing the model's training, architecture, and performance. The inference code used to run the model was publicly released under the open-source GPLv3 license. Access to the model's weights was managed by an application process, with access to be granted "on a case-by-case basis to academic researchers; those affiliated with

organizations in government, civil society, and academia; and industry research laboratories around the world". Llama was trained on only publicly available information, and was trained at various model sizes, with the intention to make it more accessible to different hardware. The model was exclusively a foundation model, although the paper contained examples of instruction fine-tuned versions of the model. Meta AI reported the 13B parameter model performance on most NLP benchmarks exceeded that of the much larger GPT-3 (with 175B parameters), and the largest 65B model was competitive with state of the art models such as PaLM and Chinchilla.

5.

Llama ("Large Language Model Meta AI" serving as a backronym) is a family of large language models (LLMs) released by Meta AI starting in February 2023. Llama models come in different sizes, ranging from 1 billion to 2 trillion parameters. Initially only a foundation model, starting with Llama 2, Meta AI released instruction fine-tuned versions alongside foundation models. Model weights for the first version of Llama were only available to researchers on a case-by-case basis, under a non-commercial license. Unauthorized copies of the first model were shared via BitTorrent. Subsequent versions of Llama were made accessible outside academia and released under licenses that permitted some commercial use. Alongside the release of Llama 3, Meta rolled out Meta AI, an AI assistant built on Llama. Meta AI has a dedicated website and is available on Facebook and WhatsApp. The latest version is Llama 4, released in April 2025. In April 2026, Meta Superintelligence Labs released Muse Spark as a replacement for Llama.

6.

Comparison of models For the training cost column, only the largest model's cost is written by default. For example, "21,000" is the training cost of Llama 2 69B in units of petaFLOPS-day. Also, 1 petaFLOPS-day = 1 petaFLOP/sec $\times$ 1 day = 8.64E19 FLOP. "T" means "trillion" and "B" means "billion". The following table lists the main model versions of Llama, describing the significant changes included with each version:

7.

Webpages scraped by CommonCrawl Open-source repositories of source code from GitHub Wikipedia in 20 languages Public domain books from Project Gutenberg Books3 books dataset The LaTeX source code for scientific papers uploaded to ArXiv Questions and answers from Stack Exchange websites In April 2023, Together AI launched a project named RedPajama to reproduce and distribute an open-source version of the Llama dataset, initially containing approximately 1.2 trillion tokens. Llama 2 foundational models were trained on a data set with 2 trillion tokens. This data set was curated to remove Web sites that often disclose personal data of people. It also upsamples sources considered trustworthy. Llama 2 - Chat was additionally fine-tuned on 27,540 prompt-response pairs created for this project, which performed better than larger but lower-quality third-party datasets. For AI alignment, reinforcement learning with human feedback (RLHF) was used with a combination of 1,418,091 Meta examples and seven smaller datasets. The average dialog depth was 3.9 in the Meta examples, 3.0 for Anthropic Helpful and Anthropic Harmless sets, and 1.0 for five other sets, including OpenAI Summarize, StackExchange, etc. Llama 3 consists of mainly English data, with over 5% in over 30 other languages. Its dataset was filtered by a text-quality classifier, and the classifier was trained by text synthesized by Llama 2. In a lawsuit against Meta Platforms brought by authors including Richard Kadrey, Sarah Silverman, and Ta-Nehisi Coates, CEO Mark Zuckerberg was alleged

to have authorized the use of copyrighted content from Library Genesis to train Llama AI models and conceal its actions by removing copyright markers from the data.